

Performance Analysis

Models Evaluated:

- Random Forest (RF)
- LSTM-based Deep Learning Model

1. Accuracy and Generalization

Model	Training Accuracy	Validation Accuracy	Observations
Random Forest	~82%	76.07%	Slight overfitting
LSTM	90.6% (Epoch 4)	84.71%	Excellent generalization

- LSTM performs significantly better in capturing sentiment and language structure.
- Random Forest struggles with sequential and contextual nuances in text data.

2. Training & Inference Time

Model	Training Time	Inference Speed	Notes
Random Forest	~6.7 min (400s)	Fast (<1s/batch)	Efficient for fast predictions
LSTM	~8.5 min total	Slower	Requires batching and benefits from GPUs

3. Memory & Compute Resources

Aspect	Random Forest	LSTM Model
RAM Usage	Low	High

Disk Size	Small (.joblib)	Medium (.h5 + tokenizer)
Hardware Need	CPU sufficient	GPU recommended
Scalability	High	Scalable with TF Serving or ONNX

4. Stability & Ease of Deployment

Factor	Random Forest	LSTM Model
Hyperparameter Sensitivity	Low	Medium-High
Reproducibility	High	Medium (requires seed control)
Packaging/Serving	Easy (joblib)	Moderate (architecture + weights)
Real-time Capability	High	Moderate

Final Model Selection: LSTM-based Model

Justification:

- **Higher accuracy (84.7%)** vs Random Forest (76.07%).
- **Better F1-score and recall**, especially for minority classes.
- Effective at capturing word order, semantics, and context in text.
- Can be improved further with attention or transformer layers.

Discarded: Random Forest:

- Simpler, but poor at capturing sequential dependencies in text data.
- Lower accuracy and recall, especially for minority classes.
- Better for baseline or low-resource environments only.

Model Evaluation Report

Tuning Summary:

- **Random Forest:**
 - Best Parameters: `n_estimators=100`, `max_depth=15`, `min_samples_split=5`, `min_samples_leaf=2`, `max_features='sqrt'`.
 - Took ~400s to tune.
- **LSTM:**
 - 10 epochs training, with best performance at Epoch 2-4.
 - Optimized with dropout and validation monitoring.
 - Saved model to .h5, with tokenizer.

Rationale for Final Model:

- **LSTM offers best trade-off** between accuracy and scalability.
- Suitable for production deployment with text-heavy applications.
- Can be extended using modern NLP techniques (e.g., transformers).

Models Discarded:

- Random Forest due to:
 - Lower generalization capacity.
 - Inability to leverage embeddings or contextual features.
- (Assumed) Other classical ML models omitted due to even worse performance.